look. The expectation value is itself a real number. Every real number is also an operator, namely an operator proportional to the identity or unit operator *I*. To make the meaning clear, we can write $\bar{\mathbf{A}}$ in a more complete form:

$$\bar{\mathbf{A}} = \mathbf{A} - \langle \mathbf{A} \rangle I.$$

The probability distribution for $\bar{\mathbf{A}}$ is exactly the same as the distribution for **A** except that it is shifted so that the average of $\bar{\mathbf{A}}$ is zero. The eigenvectors of $\bar{\mathbf{A}}$ are the same as those of **A** and the eigenvalues are just shifted so that their average is zero as well. In other words, the eigenvalues of $\bar{\mathbf{A}}$ are

$$\bar{a} = a - \langle \mathbf{A} \rangle.$$

The square of the uncertainty (or standard deviation) of **A**, which we call $(\Delta \mathbf{A})^2$, is defined by

$$(\Delta \mathbf{A})^2 = \sum_a \bar{a}^2 P(a) \qquad (5.3)$$

or

$$(\Delta \mathbf{A})^2 = \sum_a (a - \langle \mathbf{A} \rangle)^2 P(a). \qquad (5.4)$$

This may also be written as

$$(\Delta \mathbf{A})^2 = \langle \Psi | \bar{\mathbf{A}}^2 | \Psi \rangle.$$

If the expectation value of **A** is zero, then the uncertainty ΔA takes the simpler form

$$(\Delta \mathbf{A})^2 = \langle \Psi | \mathbf{A}^2 | \Psi \rangle.$$